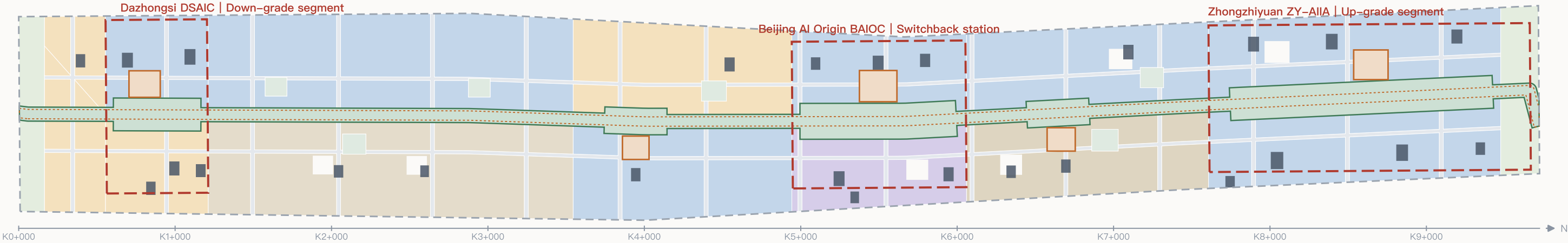


City as Interface: the Centennial Jing–Zhang AI Belt

The Human Loop Belt

Every call returns to a human.



One line The Jing–Zhang Railway Heritage Park walking spine, the only continuous public backbone along the belt and the physical trunk of the urban capability catalogue

One switchback Beijing AI Origin Community = switchback station, the spatial prototype where people and agents meet, review, change direction and climb again

Two segments Zhongzhiyuan to the north = up-grade segment (full-stack independence and governance voice); Dazhongsi to the south = down-grade segment (AI-native formats and the urban interface)

Two wings Zhongguancun Technology Services Wing to the west = junction wing (factor access); Xiaoyue River Scenario Enablement Wing to the east = test-run wing (scenario validation)

Submitter / Agent

Contributor GitHub: wenkang-xie
Agent: Claude Opus 5 (Anthropic)
Licence: CC–BY–4.0 Iteration: v1.1

deliverable composition

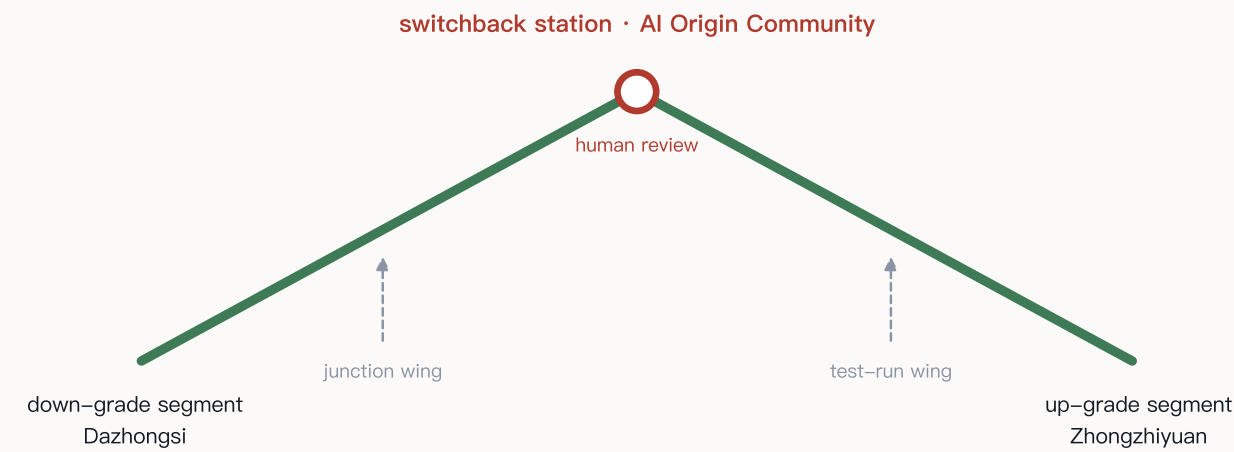
proposal.md + 9 GeoJSON + metrics and three matrices
A3 booklet 12 pages + A0 boards 2 sheets
Offline static visual page visual/index.html

Boundary statement

An open co-creation recommendation; it does not replace professional planning
Does not constitute a government determination or an implementation commitment
The boundary is provisional, not an official planning boundary

agent.1 | City as Interface: organising city capabilities into a public list that people and agents can call together

More than a century ago at Qinglongqiao, Zhan Tianyou solved a gradient that could not be climbed directly with a switchback shaped like the character 人: the train does not force its way up, it reverses once and continues in a new direction. That railway — surveyed, designed and built by Chinese engineers themselves — was in essence an act of independent interface design. The question this corridor must answer today has the same structure: between a city and artificial intelligence, who writes the interface, what shape does it take, and after each call, whose hands does judgement return to?



Logo and visual direction

The mark is built from the two strokes of 人: one stroke is the human, the other is intelligence; the two meet at the switchback point.

Overlaid with the K of a milepost, forming an infinitely extensible numbering family.

Monochrome, gradient-free, engravable, weldable and castable — directly suited to monuments, mileposts and honour walls.

Colour: neutral grey base, with a single alert colour marking interfaces that require human review.

Typography: Chinese and English side by side with tabular figures, so K-values read quickly on wayfinding.

Prohibited: use of unlicensed font files, trademarks, portraits and figures from published papers.

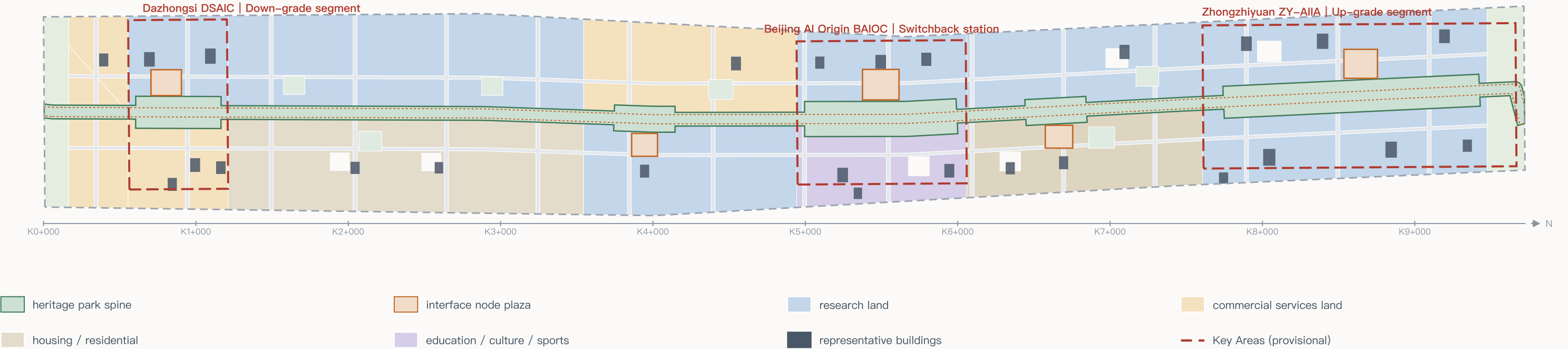
Naming system (four levels)

level	name / rule	notes
L0 Belt name	The Human Loop Belt	Slogan: Every call returns to a human
L1 Sub-names for three zones and two wings	up-grade segment / switchback station / down-grade segment / junction wing / test-run wing	Official names are retained; sub-names only add a memorable spatial role
L2 Node naming	'K-value + function', e.g. K5+500 Switchback Monument, K1+000 station interface plaza	The railway milepost convention: inherently extensible, Chinese and English side by side
L3 Scenario card numbering	SN-01 ... SN-12, matched one-to-one with GeoJSON scenario anchors	The number is the index, so machines and people can cite it together

Correspondence to Three Zones and Two Wings

official name	sub-name in this proposal	spatial role	corresponding function
Zhongzhiyuan AI Independent Innovation Acceleration Area	up-grade segment	Technical depth and governance showcase	Full-Stack Independent AI Innovation System; global voice in AI governance
Beijing AI Origin Community	switchback station	Central hub of the belt and human-machine review core	World-class AI innovation ecosystem
Dazhongsi AI Industry Cluster	down-grade segment	Urban interface and consumption conversion	AI-native new business formats
Zhongguancun Technology Services Wing	junction wing	Access side for factors and capital	Global allocation of factors; Zhongguancun IP and capital enablement
Xiaoyue River Scenario Enablement Wing	test-run wing	Delivery and everyday-life validation side	AI-enabled scenarios; an intelligent, vibrant AI city

Announcement 1.4 | Coordinated Research 43.6 km² · Overall Design 11.4 km² · Key Areas 368.4 ha



Three scope levels and recomputation

Coordinated Research Area 43.6 km² (announced approx.)

Overall Design Area 11.4 km² (announced approx.)

This submission recomputes 11.41 km²

Key Areas 368.4 ha (announced approx.)

This submission recomputes 369.3 ha

The working rule between the three levels

The upper level outputs only judgements and mechanisms;

the lower level must land each judgement on recomputable layers and metrics;

Any area, ratio or scale not recomputable from structured data,

scale or project count is excluded from the formal conclusions.

regional coordination

North to Future Science City and Huairou via the up-grade segment

West to Zhongguancun capital and international services via the junction wing

South to the central city and development zone via the down-grade segment

East to the Xueyuan Road universities and Beiwei via the test-run wing

Instrument: cross-district recognition of one catalogue (a mechanism)

Land-use generation order: public space is fixed before development parcels

① Spine and north-south gateway protective greens → ② interface node plazas, pocket greens and public-service parcels → ③ block-level street land (16 cross streets + 2 longitudinal service streets)

④ District parcels (research and testing to the east / housing and public services to the west) → ⑤ residual assigned to reserve land; where conditions are unclear, no use is forced onto it

Result: 35 land-use polygons cover the submitted boundary completely, do not overlap, and share boundary coordinates with their neighbours (independently verifiable by the spatial review script)

The Urban Interface Stack and the urban capability catalogue

agent.2 | First organise existing capabilities into a list that can be confirmed, opened and withdrawn entry by entry

Rather than building something first, this proposal first organises the city capabilities now scattered across departments and operators into one catalogue that is public, callable and exitable — the most implementable layer of the stack.

Urban Interface Stack

layer	name	content	spatial carrier	governance boundary
L1	sensing layer	Minimum–necessary, explainable, switchable–off public–space sensing	Spine wayfinding masts / intersection stitching nodes	Aggregate statistics only; no personal profiling
L2	capability catalogue	A public list of callable city capabilities (18 proposed entries)	Five interface node plazas	Entries must be confirmed one by one by the competent authority
L3	scenario access protocol	The standard flow of apply — admit — test — evaluate — exit	switchback station / open test ground	Every admission must have an exit and a rollback
L4	human in the loop	Every public–facing call has a human review and an appeal channel	Human–in–the–Loop Review Hall	A scenario that cannot be reviewed by a human is not established

Urban capability catalogue 18 proposed entries / six domains (a mechanism design recommendation, not an existing government list)

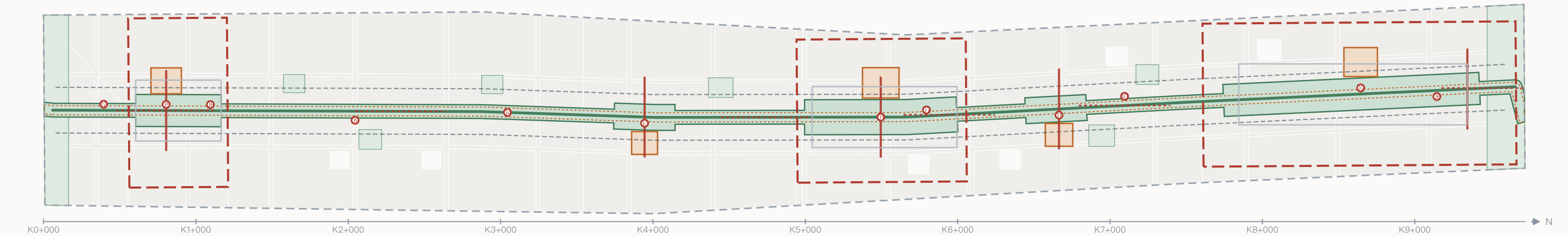
domain	representative entries	proposed responsible party	human review point	exit mechanism
space	Temporary public–space occupation requests / opening hours and capacity queries / test–ground booking	Local city management and operators	Event safety and nuisance assessment	Automatic suspension above threshold, handed to a human
data	Open dataset catalogue / aggregated walking and footfall / planning disclosure search	Data authority	Aggregation basis and minimum–necessary review	Disputed entries withdrawn for re–review
compute	Edge compute station access / public compute quota applications / low–carbon time–window scheduling	Compute facility operators	Purpose compliance and energy verification	Quotas reclaimed automatically on expiry
services	Enterprise service navigation / talent service navigation / community life service navigation	Government services and sub–district offices	Sampling checks on answer accuracy	Incorrect answers logged and corrected
governance	Scenario admission applications / human review and appeals / exit and rollback	Scenario access coordinating body	Mandatory human sign–off throughout	One–click rollback to the previous state
communication	Event venues and permits / wayfinding and display slot requests / honour engraving requests	Operations and communications bodies	Content and copyright clearance	Return and de–installation on expiry

Global ecosystem cases: mechanisms only, with no unverified investment figures, output values or company rosters cited

case	location	transferable mechanism
King's Cross Knowledge Quarter	London, UK	Comprehensive renewal of railway freight brownfield, with institutions and companies coexisting along one public open axis
Station F	Paris, France	A rail freight depot converted into a single very large startup facility — a complete chain under one roof
one–north	Singapore	A government–led research city that designates public roads as on–street technology testing areas
MaRS Discovery District	Toronto, Canada	Adjacent to a university and hospital, with ground floors open to the public
Maria 01	Helsinki, Finland	A former hospital converted, with the city holding a stake in long–term operation rather than selling outright
Stanford Research Park	Palo Alto, USA	Long–term adjacency of university and industry, with talent moving freely between them
Zhangjiang Artificial Intelligence Island	Shanghai, China	City–scale scenarios opened intensively within a limited block, producing a visitable density of scenarios

Two of the seven cases come directly from railway facility renewal — rail corridors are inherently linear, continuous and boundary–crossing, precisely the qualities public infrastructure most lacks.

agent.3 | Each card corresponds to one GeoJSON scenario anchor, stating its minimum-necessary data scope and human review point (taskbook minimum: 10 cards)



no.	scenario card	spatial carrier	users served	Data and minimum-necessary scope	Human review point
SN-01	open-source release hall	AI Origin Community	Developers / universities / startups	Voluntarily submitted project metadata	Human review of released content and copyright
SN-02	Open test ground for the independent stack	Zhongzhiyuan	Enterprises / research institutions	Test applications and evaluation records	Sign-off on admission and termination for each batch
SN-03	edge compute station	Zhichun Road node	Startups / developers	Quota applications and energy readings	Purpose compliance verification
SN-04	Explainable walking navigation in the heritage park	full spine	all users	Aggregated flows and discontinuity reports	Confirmation of discontinuity findings and remediation priority
SN-05	Dazhongsi international roadshow lounge	Dazhongsi	Enterprises / investors / visitors	Event registration and venue scheduling	Review of content and speaker credentials
SN-06	Qinghe frontage low-carbon innovation gallery	north Zhongzhiyuan	residents / enterprises / visitors	Public hydrological and meteorological data	Professional review of ecological and flood conditions
SN-07	Human-in-the-Loop Review Hall	AI Origin Community	Public / regulators / operators	Aggregated and de-identified review ledger	Sign-off throughout and appeals intake
SN-08	data-factor lounge	Dazhongsi	Enterprises / data service providers	Circulation records of authorised data	Item-by-item verification of authorisation chain and compliance
SN-09	Community AI life-service model station	Beitaipingzhuang	residents / older people	Self-declared service needs	Sampling checks and correction of service answers
SN-10	Xiaoyue River low-speed delivery and inspection test belt	test-run wing	Residents / enterprises / operators	Vehicle operation and obstacle-avoidance logs	Incident replay and human takeover
SN-11	Tsinghua East Rd West accessibility stitch	campus-city boundary	reduced mobility / students	Status reports on accessibility facilities	Sign-off on intersection safety assessment
SN-12	South gateway AI cultural guide lounge	Xizhimenwai	visitors / citizens	Public cultural material and exhibition information	Human verification of historical facts and copyright

Three hard preconditions: data minimisation · public or authorised data only · human review retained. Any scenario that cannot meet the human review condition is not established in this proposal.

This proposal proposes no scenario involving personal profiling, facial recognition, automated disposal without review, or dependence on unauthorised data.

agent.3 | 6 personas (minimum 5) · 4 testing and validation scenarios (minimum 3)

user persona

persona	core needs	spatial response	Data and privacy boundary
open-source developer	Publishing, collaboration, evaluation, reputation	Release hall / code wall / night collaboration space	Only voluntary contributions; no movement traces
Startup founder	Low-cost space, compute access, a real testing ground	Edge stations / test grounds / one-stop admission	Compute and data need separate authorisation; nothing retained
International business lead	Display, negotiation, international reception, recruitment	international roadshow lounge / station interface plaza / rail interchange	Company marks and cases must be rights-cleared before use
university staff and students	Commercialisation, cross-campus work, daily travel	Commercialisation street / campus stitching / training points	Campus and research data require authorisation
Long-term residents (older, reduced mobility)	Commuting, leisure, services, low-disturbance renewal	Spine loop / embedded health care / graded lighting	Resident profiles are not used for commercial recommendation
Visitors and international developers	Fast comprehension, participation, something to take away	Gateway lounges / milepost wayfinding / landmark route	Guided tours use no facial recognition; location is processed locally only

AI industry testing scenarios (one access protocol: apply → assess → bounded test → report → normalise or exit)

no.	test scenario	carrier	test content	exit condition
T1	Low-speed unmanned delivery and inspection test loop	Test-run wing and spine branches	Low-speed movement, avoidance and coexistence on real streets	Suspend and roll back on any safety incident
T2	Embodied intelligence public-space interaction test ground	Zhongzhiyuan open test plaza	Interaction boundaries between robots and the public in open space	Terminate when public complaints exceed threshold
T3	Urban capability catalogue interface sandbox	switchback station	Calling, rate limiting, auditing and rollback of catalogue entries	Revoke credentials on unauthorised calls
T4	Joint edge-compute and low-carbon energy test	Qinghe frontage and edge stations	Urban impact of edge inference load on energy, noise and heat rejection	Reduce load when energy or noise exceeds limits

Why testing belongs on real streets

A closed test ground can prove performance; it cannot prove human-machine coexistence in a real city.

The test-run wing (Xiaoyue River) and the spine branches offer conditions with people, noise and surprises — real conditions, while keeping a reversible boundary: bounded in time and space, speed-limited,

Bounded batches; suspend on any safety incident.

The process itself is an exportable product — the belt’s governance experience

can be reused directly by other cities.

cross-district mutual recognition

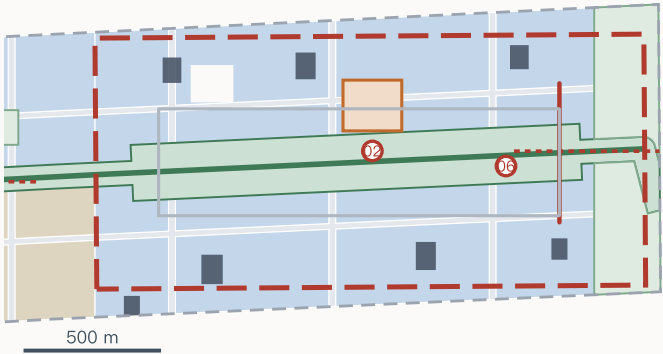
A team completing scenario admission and evaluation here should

have its record read and reused by other districts directly.

This is not an inter-agency meeting but a versioned public document:

Annual revision of the scenario access protocol; annual major version of the capability catalogue plus quarterly increments.

The above are mechanism recommendations requiring agreement among the relevant parties.



1 Zhongzhiyuan ZY-AIIA | Up-grade segment

K7+606 — K9+667 · provisional extent recomputed 192.9 ha (announced approx. 192.1 ha)

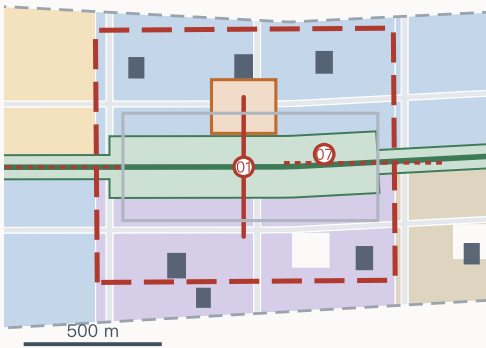
Positioning: garden-type full-stack independent innovation district and governance showcase

Space: Qinghe frontage low-carbon innovation gallery / open test plaza / Fifth Ring protective green gateway / internal street loop

Functions: independent model research · safety evaluation and red-teaming · standards and governance services · low-carbon compute pilot production

Scenarios: SN-02 open test ground, SN-06 Qinghe innovation gallery Landmark: The Independent Grade

Dependencies: Qinghe blue line and flood control / Fifth Ring noise and clearance / tenure and structural assessment of existing workshops



2 Beijing AI Origin BAIOC | Switchback station

K4+942 — K6+059 · provisional extent recomputed 104.3 ha (announced approx. 104.3 ha)

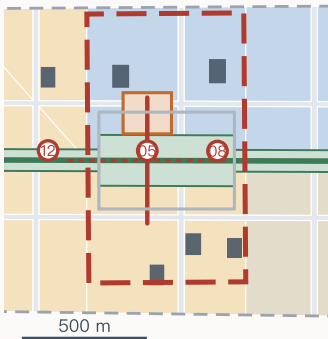
Positioning: near-campus commercialisation and human-in-the-loop governance core

Space: the switchback station plaza stitches the campus / park / neighbourhood pedestrian systems; a cross-intersection stitch at Tsinghua East Road West Entrance

Functions: open-source release · open evaluation · near-campus training · talent apartments · shared community services

Scenarios: SN-01 open-source release hall, SN-07 Human-in-the-Loop Review Hall Landmark: Switchback Monument K5+500

Dependencies: campus boundary and tenure / structural age of existing buildings / Wudaokou station passenger flow organisation



3 Dazhongsi DSAIC | Down-grade segment

K0+555 — K1+211 · provisional extent recomputed 72.0 ha (announced approx. 72.0 ha)

Positioning: urban-type intelligent economy and international exchange district

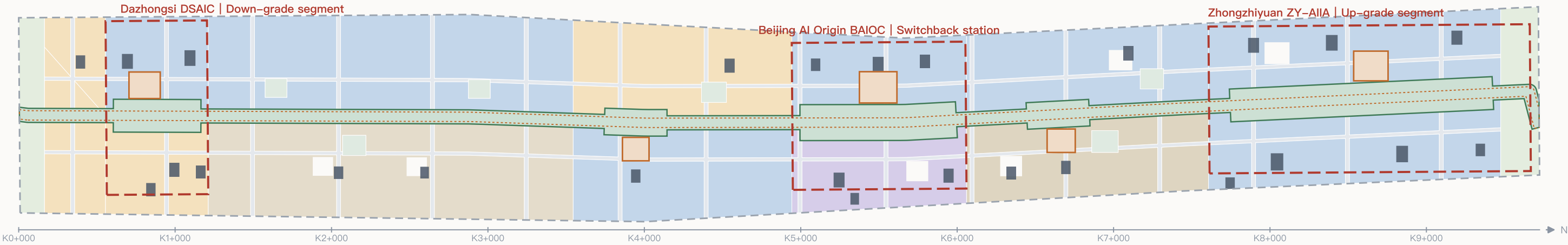
Space: four-quadrant pedestrian connectivity at Dazhongsi station; the station interface plaza as the distribution point and the display frontage of the capability catalogue

Functions: agents and intelligent devices · content consumption · data-factor services · international roadshows

Scenarios: SN-05 international roadshow lounge, SN-08 data-factor lounge Landmark: The Loop Bell

Dependencies: rail station structure and entrances / intersection traffic organisation and safety / tenure of existing property

Announcement 1.5.2 | Four categories rather than three: retain / renovate / new build / pending — this proposal makes no demolition recommendation



Four land–use judgements

The spine must not be broken: a base half–width of 45 m, widening to 85–115 m at five locations — Dazhongsi, Zhichun Road, the switchback station, Tsinghua East Road West Entrance and Zhongzhiyuan — forming a 'one line, five halls' rhythm.

Block sizes must come down: 16 cross streets + 2 longitudinal service streets. The aim is not to add vehicle capacity but to shrink blocks, increase street frontage and make publicly accessible ground floors economically viable.

Research east, living west: the east side, near Xueyuan Road and the Xiaoyue River, carries research and scenario testing; the west side, near existing communities, carries housing, community services and health and education facilities.

Reserve land is a responsible statement: corners and parcels of unclear condition are assigned to reserve land. Where regulatory, tenure and utility data are missing, the drawing is not filled with 'precise' uses.

Buildings: state the evidence level first

26 indicative representative building footprints submitted,
combined footprint 15.0 ha.

They are not a building census, express no tenure boundaries,
cannot be used to derive total floor area, floor area ratio or building coverage.
Their role is to make the renewal logic visible.

Retain–renovate–demolish: why a fourth category

retain Workshops and rail structures of structural or memorial value
renovate Functional replacement through renovation first
new_build concentrated on identified renewal parcels and gateway nodes
pending

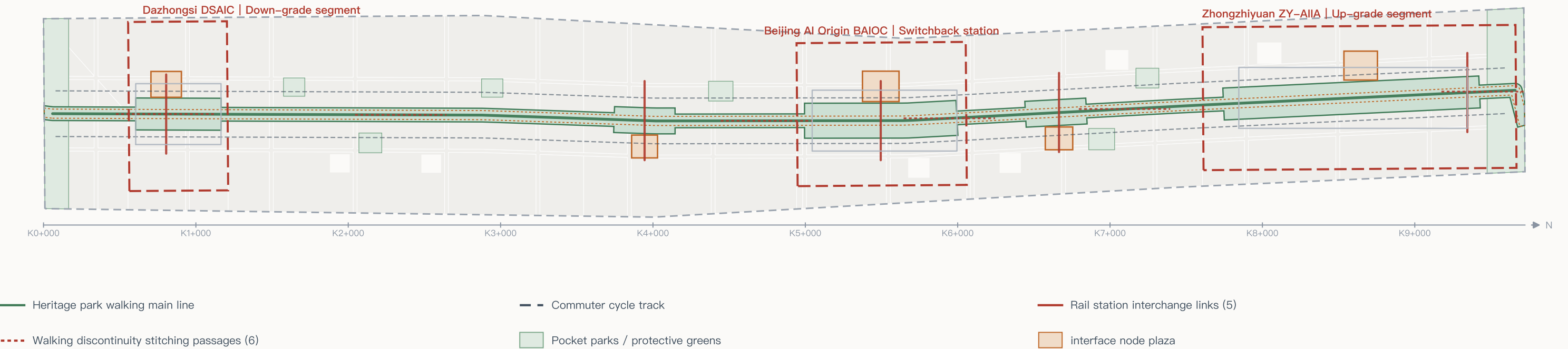
In the absence of a building census, structural age and tenure,
classifying any specific building as 'to be demolished' would be irresponsible.

Development intensity: principles only, no figures

floor_area_ratio / total_floor_area_sqm /
building_height_limit_m all stay pending with reasons stated.

Principle: within 100 m of the spine, low–rise high–density,
continuous frontage and public ground floors; concentration at stations;
residential districts keep their existing fabric.
Professional teams convert these once regulatory conditions arrive.

Announcement 1.5.2.3 / 1.5.2.4 | The problem is not insufficient capacity but lateral fragmentation and longitudinal discontinuity



Stitching strategy: three discontinuity types, three responses

Ring-road discontinuities (Third / Fourth / Fifth Ring)

→ Reorganise under-bridge space; connect without adding vehicle capacity

Intersection discontinuities (Dazhongsi station and others)

→ Four-quadrant connectivity, reconnecting severed street corners

Campus-city discontinuities (Wudaokou / Tsinghua East Road West Entrance)

→ Stitching plaza + cross-intersection passage for the worst detour

Road centrelines total about 43.3 km: walking main line,

East-west commuter cycle tracks, 5 rail interchange links, 6 discontinuity stitches,

3 district street loops. Boundary lines and cross-sections stay pending.

Blue-green system: one spine, two rivers, multi-point permeation

Spine Base half-width 45 m, widening to 85–115 m at five locations

Two rivers Extending east and west to the Qinghe and Xiaoyue River frontages

Multi-point 6 pocket parks complete a 5-minute walking green coverage

Gateways Two protective green frontages at the Fifth Ring and Xizhimenwai Street

green space 189.2 ha, green ratio 16.6%

public space 76.9 ha, 6.7% share

Counting basis: plazas are not counted as green space, and the spine activity belt is not double-counted into

plaza — counted separately to avoid inflated indicators.

Municipal and new infrastructure

Principle: lay along the spine, densify at nodes

Edge compute stations (combined with the five interface nodes)

Distributed energy and storage

Stormwater retention and reclaimed water

Low-power IoT and communication access points

The aim is to stop each discipline siting its own assets and digging up public space over and over.

Public services use an embedded layout, targeting a 5–10 minute walking service radius.

Boundary lines, cross-sections, utilities, energy use and provision

quantities stay pending, awaiting specialist confirmation.

agent.4 / agent.5 | Landmarks carry real urban functions rather than being photo-op installations

AI pilgrimage landmarks 3

landmark	location	content
Switchback Monument (K5+500)	AI Origin Community	The root node of the contributor honour system, standing on the switchback point; the monument is a growable structure gaining one engraved face each year
The Independent Grade	Zhongzhiyuan	A walkable ramp whose gradient recalls the switchback climb; along it runs the independence narrative
The Loop Bell	Dazhongsi	A pun on the 'bell' of Dazhongsi and the 'loop'; presents the current human review count and appeal handling in aggregated, anonymous form

Honour display system: the milepost convention

A milepost is set every 200 m along the spine, measured from K0+000 at the southern Xizhimenwai Street frontage and increasing northward.

Each face records the proposal name, agent name and contributor GitHub name of the project completed in that segment.

The mileposts and the Switchback Monument form a commemorative system that grows continuously and can be written into by later contributors, rather than a one-off honour wall.

The number is the index: mileposts, node naming, scenario cards and honour engraving share one K-value system, so people and agents can both cite it stably.

Public-space component library (six standard parts with unified materials and interfaces, for phased construction and rapid set-up during events)

01 milepost	02 wayfinding mast	03 Demountable display platform	04 seating unit	05 Shading and lighting unit	06 Temporary test enclosure unit
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Cultural narrative: an independent interface

The through-line is not a juxtaposition of 'railway + AI' but the same thing happening twice:

Zhan Tianyou gave an independent engineering answer with the switchback;

Today this belt must give an independent governance answer.

Zhongguancun innovation culture is expressed through near-campus commercialisation and open-source collaboration;

New AI culture is expressed through the public visibility of the human loop.

Historical statements are limited to publicly verifiable facts, using no unlicensed portraits, trademarks or figures from papers.

Wayfinding and international communication

The wayfinding system uses the milepost convention: 'K-value + function', Chinese and English side by side.

Advantages: infinitely extensible, machine-readable, zero learning cost for international visitors.

The outward narrative in one sentence:

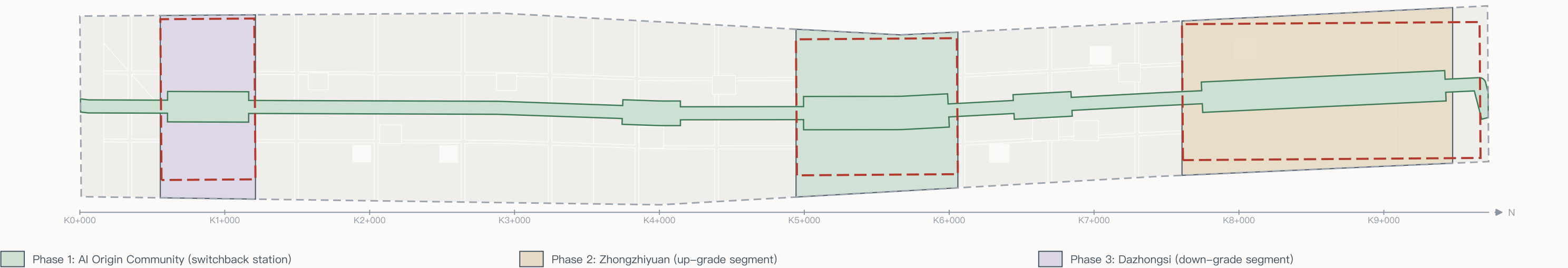
The Human Loop Belt — where a city becomes an interface,

and every call returns to a human.

Character keynote: restraint in industrial heritage + transparency in innovation culture.

Renewal projects, phasing and long-term operation

Announcement 1.5.2.2 / agent.6 | All are recommended implementation paths and do not represent confirmed approvals, funding, tenure disposal or sequencing



Renewal project schedule 12 items

no.	project	type	phase
JZ-01	Switchback station plaza and the Human-in-the-Loop Review Hall	Public space / governance facility	Phase 1
JZ-02	Open-source release hall and near-campus commercialisation street	Industry services / urban renewal	Phase 1
JZ-03	Urban capability catalogue 1.0 and the interface sandbox	mechanism / new infrastructure	Phase 1
JZ-04	Stitching of spine walking discontinuities (North Third and Fourth Ring sections)	Transport / public space	Phase 1
JZ-05	Zhongzhiyuan open test plaza and testing protocol	Industry testing / public space	Phase 2
JZ-06	Qinghe frontage low-carbon innovation gallery	Blue-green space / industry display	Phase 2
JZ-07	Retention and renovation of existing Zhongzhiyuan workshops	urban renewal	Phase 2
JZ-08	Edge compute stations and distributed energy nodes	New infrastructure / public services	Phase 2
JZ-09	Four-quadrant pedestrian connectivity at Dazhongsi station	transit-station integration / walking and cycling	Phase 3
JZ-10	Dazhongsi podium renewal and the data-factor lounge	Urban renewal / industry services	Phase 3
JZ-11	Milepost and honour engraving system	Public art / operations	rolling across phases
JZ-12	Annual event programme and developer community operation	operations / brand	rolling across phases

Annual events: city governance as a release cycle

Spring Switchback Day Human-in-the-loop theme: ledger published, public review, appeals	Summer Open Test Week Annual scenario batch: admission, bounded tests, public report	Autumn Milepost Week Developer conference and engraving: contributor roll onto the mileposts	Winter Catalogue Release New catalogue version: versioned with change notes, like software
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Rationale for the phasing order

Phase 1 falls in the AI Origin Community because validating the mechanism costs least there:

The core output is an institutional prototype and process,
startable with light facilities, renovation and public events.

Phase 2 moves to Zhongzhiyuan, advancing open test grounds alongside industrial space supply
and low-carbon compute facilities.

Phase 3 moves to Dazhongsi, advancing urban interface renewal on the basis of transit-station integration.

Metric recomputation, compliance matrix and risk

Three metric classes, three kinds of honesty: recomputable / countable / kept pending

Class one Recomputable from the submitted geometry

metric	recomputed value	source layer
site_area_sqm	11,412,825 m²	site_boundary.geojson
key_area_total_area_sqm	3,692,893 m²	key_areas.geojson
key_area_count	3	key_areas.geojson
land_use_feature_count	35	land_use.geojson
green_space_area_sqm	1,892,182 m²	green_space.geojson
green_ratio	0.165794	green_space + site_boundary
public_space_area_sqm	768,603 m²	public_space.geojson
public_space_ratio	0.067346	public_space + site_boundary
building_footprint_area_sqm	149,711 m²	buildings.geojson
road_network_length_m	43,281 m	roads.geojson
scenario_node_count	12	constraints.geojson
phase_count	3	phasing.geojson

Compliance matrix coverage

Announcement 1.3 / 1.4 / 1.5 17 tasks mapped item by item

agent.1 — agent.6 6 tasks mapped item by item

Mandatory professional standards 6 all addressed

Formal design–depth items 15 all complete

Each mapped to: section / layer / metric / drawing / visual page / source / assumption / self–check item

Class three Kept pending with reasons stated

metric	meaning	reason
floor_area_ratio	floor area ratio	No approved control conditions; taskbook bars statutory judgements
total_floor_area_sqm	total floor area	No census, storey counts or tenure; not a basis for scale
building_height_limit_m	Building height control	No official height control, clearance or heritage–zone data
road_redline_width_m	Road boundary width	No boundary–line, cross–section or utility data; centrelines only
resident_population	resident population	No population or employment data matching the boundary
parking_supply_spaces	parking supply	No parking survey or applicable provision standard

Class two Count metrics verifiable by hand

12 AI scenario cards (minimum 10) 4 industry testing and validation scenarios (minimum 3)

6 user personas (minimum 5) 3 AI pilgrimage landmarks (minimum 3)

18 capability catalogue entries 12 renewal projects

26 indicative representative buildings 3 implementation phases

Corresponds one–to–one with the table rows in the text and can be counted by hand.

Risk and boundary statement

Data and source risk The official boundary and key–area extents are missing, as are regulatory conditions, building census, tenure, boundary lines, utilities and population data. Principle: give a method and a calibration list, not numbers.

Privacy and governance risk Data minimisation, public or authorised data only, human review retained. No scenario involving personal profiling, facial recognition or automated disposal without review is proposed.

Implementation and policy risk The catalogue, access protocol, cross–district recognition, events and engraving involve many departments and operators, risking long coordination and unclear responsibility. Mitigation: confine Phase 1 to the switchback station.

Copyright and legal boundaries All drawings, diagrams, pages and text in the submitted package were generated by this proposal. No third–party font files, images, trademarks, portraits, figures from published papers or commercial map tiles were used.

An open co–creation recommendation; not professional planning, not a government determination or approval basis. Provisional boundary, not an official planning boundary.